

CAUSAL DISSECTION OF SCIENTIFIC AGENTS: BREAKS FROM EVIDENCE TO ACTION

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ABSTRACT

Experimental success does not reveal whether a scientific agent learned the right model. We use ChemWorld as a controlled causal probe: physics, interface, resources, and stochastic identity stay fixed while the initial model is opaque, aligned, or misspecified at entity, parametric, or structural loci. Across 135 public sessions, search changed and prediction error fell in every arm and locus, yet all three selective-correction criteria failed. Matched structural evidence reduced mean error from 0.2255–0.3392 to 0.0060–0.0074, but DeepSeek and GPT recovered the registered 1.75 power law in 0/5 misspecified worlds; the same break survived a lower-reasoning ablation. On matched 30-cell identifiable-law surfaces, joint recovery was 0/30 for DeepSeek and 5/30 for Codex, with no registered action gain in either model. A 135-cell Codex replication lowered executable-law MAE from 0.237 to 0.175, yet blind gain remained approximately zero in both models. An oracle-free four-condition successor found no stable learned-law benefit; autonomy was directionally favourable but intervals crossed zero and yoked failures were substantial. Only 11/42 fresh action readouts selected Top-1. Finally, a ranking control passed 7/8 fresh units while selecting Top-1 in 1/8, whereas another failed the rank gate but selected Top-1 with zero regret. Numerical revision, structural identification, executable compression, decision transfer, and evaluator validity are distinct transitions rather than one score.

1 INTRODUCTION

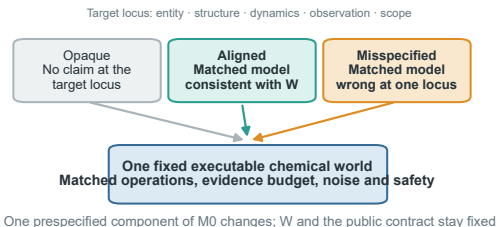
Autonomous experimental agents are increasingly judged by whether they discover high-performing molecules or reactions. Yet an endpoint benchmark cannot distinguish whether an agent inherited a fortunate prior, navigated a productive search corridor, adopted a phenomenological interpolation, or genuinely revised its scientific understanding from evidence. We call this ambiguity endpoint underdetermination. The relevant question is not only whether an agent finds a good experiment, but what that experiment changes in its model of the world.

This distinction matters as language-model agents plan syntheses, call chemistry tools, operate instruments, and enter self-driving laboratory workflows (Boiko et al., 2023; Bran et al., 2024; Szymanski et al., 2023; Darvish et al., 2025; Song et al., 2025; Vriza et al., 2026). Interactive environments increasingly test repeated cycles of hypothesis, intervention, and inference (Jansen et al., 2024; Gandhi et al., 2025; Duan et al., 2025; Zheng et al., 2026; Yang et al., 2026; Batzoglou, 2026). Yet pretrained knowledge, prompt-provided information, experiment selection, endpoint optimization, and verbal explanation are usually observed together. Consequently, a high score need not identify which transformation succeeded, and a low score need not identify which transformation failed.

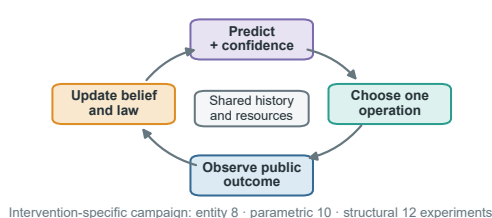
We make those transformations observable by using ChemWorld as a controlled causal probe. The same executable world, interface, resources, and stochastic identity are held fixed while its initial model is made opaque, aligned, or misspecified at a declared entity, parametric, or structural locus. Persistent agents then interact through a transactional laboratory interface. Evaluator-owned counterfactual queries, matched contradictory evidence, executable-law assays, and unseen complete ActionPlans separately measure search, prediction, structural identification, compression, and decision transfer.

Endpoint success does not reveal what the agent learned

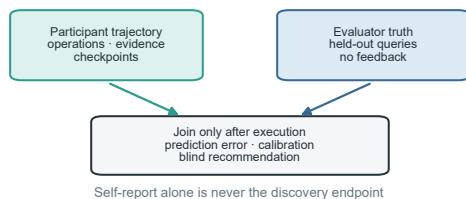
a Intervene on the initial model, not the world



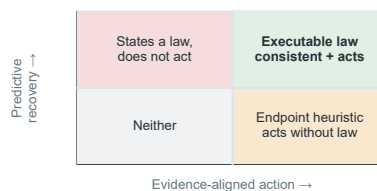
b Trace one evidence-to-action trajectory



c Score understanding after the campaign



d Locate the conversion that failed



The agent-facing initial model is manipulated under one fixed world; search, prediction, executable law and unseen-plan selection remain separate evidence channels.

Figure 1: Endpoint success underdetermines scientific competence. The same outcome can arise from a correct prior, local search, or evidence-driven model correction. ChemWorld intervenes on the initial model while holding the executable world fixed, then evaluates the transitions from evidence to predictions, executable laws, and unseen actions.

Our contribution is a causal map rather than a scalar leaderboard. Four breaks emerge: priors redirect search without selective repair; matched evidence yields numerical convergence without reliable structural recovery; executable laws can lose information present in explicit predictions; and both agent-level law adequacy and evaluator-level rank qualification decouple from action validity. This locates losses in revision, identification, compression, transfer, and evaluation.

2 ENVIRONMENT AND EVALUATION

2.1 CHEMWORLD AS A CONTROLLED CAUSAL PROBE

ChemWorld separates three objects. The executable world contains hidden state, transition and measurement laws, resources, and terminal assays. The public task exposes typed operations, instruments, observations, and an initial model. The evaluator owns held-out truth, replay, and scoring. For world w and initial-model arm a , the hidden transition process remains fixed while the supplied model $M_{0,a}$ changes. The aligned model is correct at one declared locus, the misspecified model is explicit but wrong at that locus, and the opaque arm withholds the corresponding structure. Because operations and measurements remain identical, differences can be attributed to information supplied to the complete agent–tool system rather than to different physics.

Actions are transactions, not free-form prose. The host validates a typed request, checks resources and preconditions, commits the state transition, and returns a public observation or a structured rejection. Failed actions and discards remain in the trajectory. Every submitted action, commit, rollback, measurement, assay, and resource delta is recorded for exact replay. These semantics matter: an action recommendation is scientific evidence only when the evaluator can execute the same complete plan without filling in hidden workflow choices.

2.2 FIVE-STAGE CAPABILITY CHAIN

We distinguish five downstream outcomes. **Evidence acquisition** asks which informative experiments are selected and observed. **Numerical revision** asks whether held-out counterfactual prediction error falls. **Structural identification** asks whether the agent rejects a false form and recovers the registered relation. **Executable compression** asks whether a typed law preserves the information in conditional predictions. **Decision transfer** asks whether the final state selects a good previously unseen plan. A sixth requirement, **evaluator validity**, asks whether the control and metric actually measure the decision estimand. Success at one stage is neither defined as nor assumed to imply success at the next.

At five checkpoints per session, the participant reports initial-model reliability, predictions, uncertainty, evidence references, an executable law, and the next experimental intent. The evaluator executes prespecified counterfactual query sets independently of the participant. It scores 675/675 checkpoints from 420/420 truth executions and later runs each final typed law on the same coordinates. Paired blind replay evaluates the final recommendation against the observed incumbent, while a separate longitudinal assay reveals eight outcome-hidden complete ActionPlans only after autonomous exploration has ended.

3 EXPERIMENTAL PROGRAMME

The prospective programme is layer-stratified. Entity interventions cover five task families and five independently selected public worlds per task. Parametric and structural blocks each cover two validated task families and five worlds. Every task–world cluster contains opaque, aligned, and misspecified arms, yielding 45 clusters and 135 independent persistent sessions. Campaign length is locus-specific: eight, ten, or twelve complete experiments, with five checkpoints in every session. Exploratory, validation, prospective, matched-evidence, and open-action worlds remain separated.

The primary contrast tests selective evidence-driven correction. If $E_{a,k}^{(\ell)}$ is held-out error for arm a , checkpoint k , and locus ℓ , then

$$C_\ell = \left(E_{\text{mis},0}^{(\ell)} - E_{\text{mis},K}^{(\ell)} \right) - \left(E_{\text{aligned},0}^{(\ell)} - E_{\text{aligned},K}^{(\ell)} \right).$$

Success requires greater correction in the misspecified arm, improvement of that arm, and no material deterioration of the aligned arm. Loci are decided separately; unlike intervention semantics are not pooled. Failed scientific cells stay in their scheduled denominator, with the last valid checkpoint carried forward for right-censoring and zero primary improvement assigned when the final prediction is missing. Only a pure infrastructure failure without a persisted trajectory can resume under a fixed attempt cap.

Matched-evidence sessions use cloned worlds and provide the same decisive counterevidence to every arm, separating failure to seek evidence from failure to update after seeing it. The longitudinal action matrix separately contains three tasks, five worlds, and three arms (45 scheduled cells). After 12 autonomous experiments, each agent ranks eight new plans; regret and Top-1 are primary action readouts, while complete-rank correlation and law adequacy are diagnostics. Table~1 keeps these layers and their claim boundaries explicit.

Provenance is block-specific: C2 and B3 have matched scheduled DeepSeek-high/Codex-medium surfaces; matched evidence adds complete denominators and a DeepSeek-low structural ablation; the four-condition successor retains model-specific donor eligibility. Provider-free controls and model configurations are never pooled.

4 RESULTS: FROM PRIORS TO LAWS

4.1 PRIORS SCAFFOLD SEARCH WITHOUT SELECTIVE CORRECTION

All 135 scheduled sessions produced final records. Participants completed 1,243/1,260 planned experiments; 121 sessions met operational eligibility. The denominator retains 26 discarded lifecycles,

Table 1: Executed evidence layers and claim boundaries. A completed work package may contain a retained scientific rejection; unstarted units are not silently removed.

Layer	Units	Execution	Supported role
Prospective C2	270 scheduled	DeepSeek 121 complete; Codex 126 complete	Search, prediction, law, incumbent replay
Matched evidence	75 sessions	DeepSeek high + GPT (60); B2 low (15)	Replicated numerical-structural break
Identifiable-law B3	60 scheduled	DeepSeek 17+13 failures; Codex 30	Structural recovery and action bridge
Action assays	45 cells + 360 slots	W2-50 + four conditions per model	Descriptive and failure-aware action transfer
96-query control	15 planned	Provider-free; 8 attempted	Retained pre-participant rejection
320-query control	7 exposed + 15 fresh	Provider-free; 7 pass + 1 fresh	Construction repair versus prospective rejection
Gate alignment	16 unit versions	Provider-free; 0 new execution	Rank validity versus action validity

13 resource-ledger rejections, and all right-censored cells. Every session submitted five checkpoints, providing 6,300 counterfactual predictions and 24,300 query-metric values.

The intervention reached behavior. The first complete recipe differed between aligned and misspecified cells in 45/45 matched clusters, between opaque and aligned in 45/45, and between opaque and misspecified in 44/45. This is a manipulation check rather than a causal effect estimate because there are no repeated same-arm sessions. Search continued after the first proposal: 91.2% of completed experiments used a unique recipe, 84.4% of session optima appeared after the midpoint, and 32.6% appeared in the last completed experiment.

Correct-prior utility was task dependent. In entity-level partition, aligned information produced a +0.200 best-endpoint advantage over the misspecified arm in 5/5 worlds. In structural crystallization, a +0.141 first-experiment head start narrowed to +0.055 as the disadvantaged arm explored. In structural partition, aligned and misspecified descriptions both helped relative to opaque identifiers while differing little from one another. A correct model therefore altered the search landscape without imposing one endpoint ordering.

Prediction error nevertheless fell on average in every arm at every locus. Reductions for opaque, aligned, and misspecified cells were 0.111/0.097/0.097 at the entity locus, 0.090/0.033/0.065 at the parametric locus, and 0.219/0.228/0.221 at the structural locus. The stricter selective-correction contrasts were -0.214 for entity ($p = 0.990$), +0.033 for parametric ($p = 0.079$), and -0.224 for structural ($p = 1.000$); none passed. General predictive learning did not become preferential correction of a wrong starting model (Fig.~2).

4.2 MATCHED COUNTEREVIDENCE SUPPORTS INTERPOLATION WITHOUT STRUCTURAL RECOVERY

Free discovery cannot distinguish failure to seek diagnostic evidence from failure to use it. In the parametric matched-evidence block, all five misspecified summaries rejected the supplied high-potential direction and recovered the peak-and-collapse response once the decisive profile was provided. Part of the free-discovery loss therefore lay in evidence acquisition.

The corrected structural DeepSeek assay showed a different bottleneck. After every arm received the same direct phase-process evidence, mean normalized errors fell from 0.2255, 0.2736, and 0.3392 to 0.0074, 0.0060, and 0.0071 for opaque, aligned, and misspecified cells. The misspecified-minus-aligned update contrast was positive in only 3/5 worlds (exact one-sided sign-flip $p = 0.125$), so this is numerical convergence rather than a confirmatory arm effect. Yet 0/5 misspecified public summaries recovered the registered 1.75 power law, 1/5 explicitly rejected the supplied linear form, and 5/5 adopted a saturation or endpoint model. Direct evidence supported accurate local phenomenological interpolation without structural identification (Fig.~3).

The matched GPT replication preserved this break. A-P misspecified-minus-aligned update contrasts were +0.0309 for DeepSeek and +0.0602 for GPT; structural B2 contrasts were +0.0645 and +0.0915, while misspecified exact-law recovery remained 0/5 in each model. This is a block-specific cross-model replication, not a model-superiority test.

Changing only the DeepSeek Codex reasoning effort from high to low preserved the structural break. The B2 canary passed 3/3 and formal execution completed 15/15 with no failures. All 15 post errors

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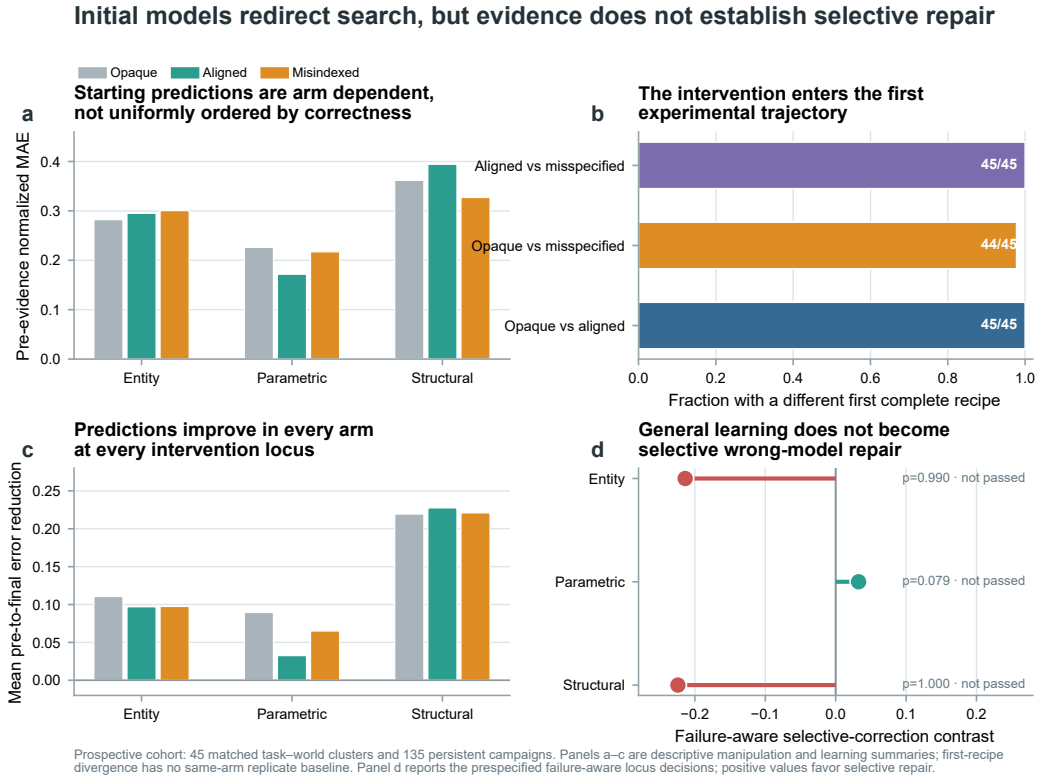


Figure 2: Initial models redirect search, but the evidence does not establish selective repair of the wrong model. Pre-evidence errors, first-recipe divergence, pre-to-final error reductions, and frozen failure-aware selective-correction contrasts are reported by intervention locus. Every arm improves on average, but no locus passes the selective-correction criterion.

were below 0.02, while the registered contrast reversed to -0.0405 (2/5 positive worlds; exact one-sided $p = 0.8125$) and misspecified exact-law recovery remained 0/5. Provider-reported reasoning output was 20.9% lower than DeepSeek high on the same block. The low A-P canary lacked terminal receipts and contributes no estimate. This is a same-harness reasoning-budget robustness result, not reasoning-off or a configuration ranking.

An independent control made the 1.75 exponent identifiable on a disjoint scoring roster. Codex completed 30/30 sessions; joint family–exponent recovery was 0/10, 5/10, and 0/10 for opaque/aligned/misspecified, and 0/18 action-eligible cells achieved gain at least 0.02. The matched DeepSeek successor retained 17 completed cells and 13 schema failures; failure-aware joint recovery and Top-1 were 0/30, versus 5/30 and 2/30 for Codex. Neither model achieved the registered useful action gain. Differential failures preclude model ranking, while the shared negative action bridge strengthens the transition-level result.

All 135 DeepSeek laws executed, but 84 lost information relative to final explicit predictions; mean law MAE was 0.237 and compression loss 0.069. Legal full-basis controls reproduced 135/135 prediction states at mean MAE 4.25×10^{-13} , localizing the gap to participant distillation rather than typed-interface capacity. Blind incumbent replay completed 726 executions for 121 cells, with recommendations better/equivalent/worse in 1/119/1.

The matched Codex surface retained 126 completed, 3 failed, and 6 right-censored cells. All locus gates again failed. Codex versus DeepSeek law MAE was 0.1753 versus 0.2371 and compression loss 0.0142 versus 0.0686, but blind gain was -0.0001 versus -0.0010. Better compression still did not improve action (Fig.~4); model differences are descriptive.

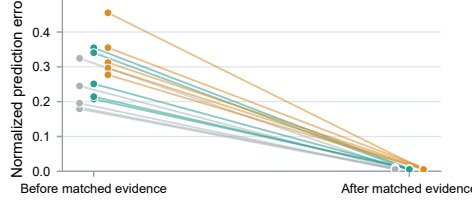
Matched evidence yields numerical convergence without reliable law-to-action recovery

45/45 structural sessions across 3 configurations

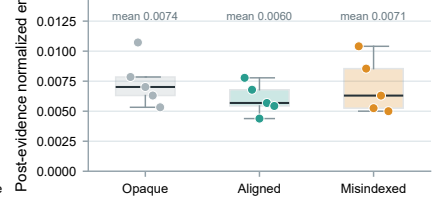
B3: 5/60 correct laws; 0 useful action gains

● Opaque ● Aligned ● Misindexed

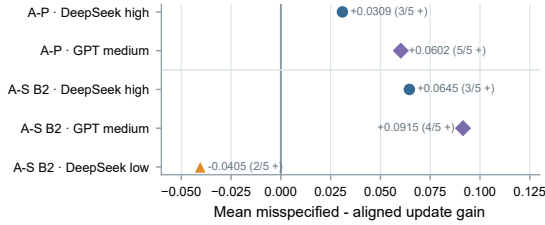
DeepSeek high: identical evidence drives all arms toward accurate predictions



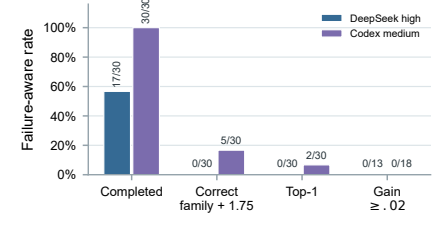
DeepSeek high: post-evidence predictions converge across initial-model conditions



Matched numerical-update contrasts across replicated configurations



Identifiable-law recovery still does not bridge to useful action



Panels a-b show the corrected DeepSeek-high B2 structural assay; panel c adds matched GPT-medium and DeepSeek-low evidence. All three B2 configurations put misspecified cells near the observed outcomes yet recover the exact 1.75 law in 0/5 worlds. Panel d reports the separate participant-identifiable B3 surface: DeepSeek has 13 retained schema failures, Codex recovers 5/30 joint laws, and neither model realizes a registered useful action gain. Five-world and model comparisons are descriptive; no capability-ranking test is performed.

Figure 3: Matched evidence yields numerical convergence without reliable law-to-action recovery. Panels a–b detail DeepSeek high. Panel c compares matched A-P DeepSeek/GPT and structural B2 DeepSeek-high/GPT-medium/DeepSeek-low contrasts. Panel d reports failure-aware completion, joint-law recovery, Top-1, and useful action gain on matched 30-cell B3 surfaces.

5 RESULTS: LAW, ACTION, AND EVALUATION

5.1 AGENT-LEVEL LAW ADEQUACY DOES NOT DETERMINE UNSEEN ACTION SELECTION

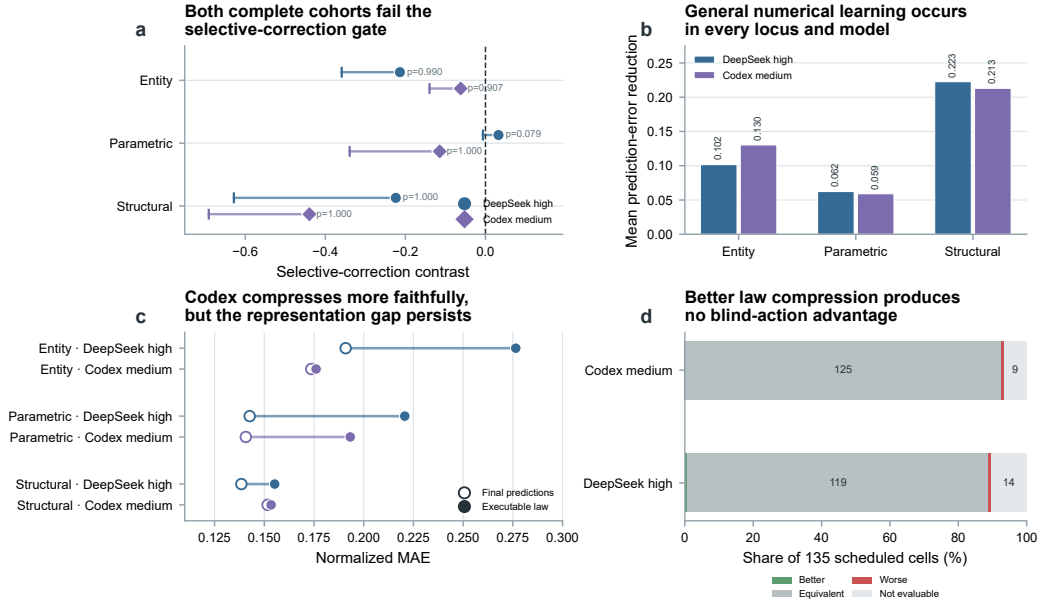
The longitudinal DeepSeek matrix produced records for 45/45 scheduled cells. Truth evaluation and exact replay both completed 240/240 plan executions. Forty-two cells were uncontaminated and eligible for action metrics; two agent-induced resource/process failures and one session interruption in crystallization remain in the scheduled denominator.

Only 11/42 eligible terminal readouts selected the true Top-1 plan. Mean selected rank was 3.31 of 8 and mean normalized regret was 0.297. The uniform-random rank 4.5 is geometric context, not a no-evidence control, so these outcomes describe post-campaign competence rather than a causal benefit of exploration. Law and action also separated: 30 cells had an inadequate law and wrong action, 11 an inadequate law but correct action, one an adequate law but wrong action, and none an adequate law and correct action. Task means ranged from rank 2.00 for reaction safety to 4.58 for crystallization, and every pairwise arm contrast changed sign under some leave-one-cluster-out omission. We therefore make no pooled arm-level claim (Fig.~5a,b).

This separation survives continuous and threshold-sensitive analysis. Pooled law MAE correlated weakly with selected rank (Spearman $\rho = -0.073$, cluster-bootstrap 95% interval $[-0.380, 0.256]$) and normalized regret ($\rho = -0.133$, $[-0.452, 0.217]$), while task-specific rank associations reversed sign (+0.524, -0.592, and -0.007). Across law-MAE thresholds from 0.05 to 0.30, the adequate subset expanded from 1 to 34 cells but contained only 0 to 9 correct actions. The four-way table is therefore not a single-cutoff artifact: law error has no stable monotonic map to unseen-action quality across tasks.

Better executable laws still do not produce better actions

Matched 135-cell C2 cohorts: law MAE 0.237 → 0.175; blind gain -0.0010 → -0.0001



DeepSeek and Codex use the same 45 task-world clusters, nine tasks and three prior arms. Panels a-b retain the registered failure-aware correction decisions; panel c is matched descriptive and separates final explicit predictions from submitted executable laws. Panel d keeps every scheduled cell. DeepSeek better/equivalent/worse/not-evaluable is 1/119/1/14 and Codex is 0/125/1/9. Model differences are not provider causal effects or a leaderboard.

Figure 4: **Better executable laws still do not produce better actions.** **a**, Registered correction estimates and lower bounds for both 135-cell C2 cohorts. **b**, Numerical learning by locus and model. **c**, Final-prediction versus executable-law MAE. **d**, Blind outcomes over every scheduled cell; model contrasts are descriptive.

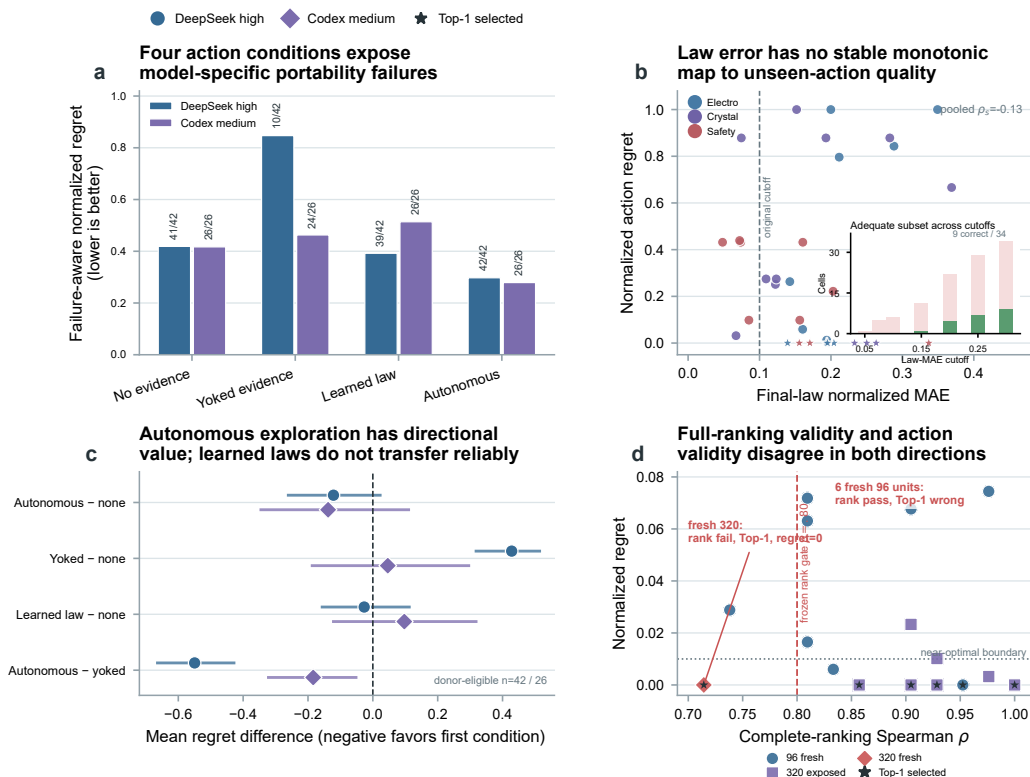
5.2 FOUR CONDITIONS SEPARATE AUTONOMY FROM ARTIFACT PORTABILITY

An independent development successor scheduled no evidence, yoked evidence, learned-law-only, and autonomous-exploration action readouts for all 45 strata per model: 180 slots each and 360 total. DeepSeek and Codex had 42 and 26 donor-eligible strata, with 26 coordinates shared. Within those model-specific populations, autonomous-minus-no-evidence regret was -0.1214 (95% cluster interval [-0.2607, 0.0225]) and -0.1379 ([-0.3452, 0.1103]); both favour autonomy but cross zero. Learned-law-minus-no-evidence was -0.0267 and +0.0970, also with intervals crossing zero. Yoked completion was only 10/42 and 24/26, so its failure-aware disadvantage cannot be interpreted as a pure experiment-selection effect. The bounded result is directional total value for autonomy without reliable transfer of the compressed law (Fig.~5a,c).

5.3 EVALUATOR-LEVEL RANKING AND DECISION QUALITY ARE DIFFERENT ESTIMANDS

A planned five-condition follow-up required an outcome-disjoint oracle law to rank eight candidates with Spearman $\rho \geq 0.80$. The first eight fresh clusters completed 896/896 truth/replays, but the eighth obtained $\rho = 0.738095$ and the wrong Top-1; the frozen stop left all 225 participant sessions unstarted. Expanding 96 queries to a 320-query grid repaired 7/7 exposed units, yet the first fresh world failed at $\rho = 0.714286$ while selecting Top-1 with zero regret. A zero-execution diagnostic then reproduced all 16 unit versions: fresh 96-query rank gates passed 7/8 but Top-1 only 1/8, whereas the fresh 320-query unit was rank-fail/action-correct. Full-order correlation and decision validity therefore disagree in both directions (Fig.~5c,d; Table~2); the historical stops remain valid, while future controls should prioritize regret and near-optimality.

Evidence, learned laws and evaluator rankings diverge from action



W2-61: 360 scheduled condition slots; donor-eligible DeepSeek/Codex strata = 42/26 and all recipient failures remain in the failure-aware estimates. W2-50: 45 scheduled, 42 eligible and 240/240 truth/replay. W2-51/W2-52 contain zero participant sessions; W2-53 reproduces 16/16 frozen unit versions. Full-ranking correlation is a secondary diagnostic, and model differences are not provider causal effects.

Figure 5: **Evidence, learned laws, and evaluator rankings diverge from action.** **a**, Failure-aware regret across four W2-61 action conditions in model-specific donor-eligible populations. **b**, W2-50 law error versus regret, with Top-1 stars and a cutoff inset. **c**, Registered W2-61 contrasts with clustered intervals; yoked failures are retained. **d**, Spearman correlation versus regret for all 16 frozen W2-53 unit versions.

6 RELATED WORK

Self-driving laboratories and chemistry agents emphasize closed-loop execution, tool use, or endpoint optimization (Felton et al., 2021; Häse et al., 2021; Boiko et al., 2023; Bran et al., 2024). Virtual laboratories and process-control environments provide scalable interaction and safety (Beeler et al., 2024; Bloor et al., 2024; Malik et al., 2026; Chen et al., 2025). Scientific-discovery benchmarks increasingly test iterative experimentation, causal inference, and transferable knowledge (Jansen et al., 2024; Gandhi et al., 2025; Duan et al., 2025; Yang et al., 2026; Batzoglou, 2026). ChemWorld complements these directions by making the initial model an experimental variable and by evaluating each transformation from evidence to action with evaluator-owned truth and exact replay. The central distinction is diagnostic: endpoint success, prediction learning, law recovery, and action quality are not interchangeable readouts.

7 DISCUSSION AND LIMITATIONS

The systems search productively and learn numerically, yet lose information at distinct transitions. Priors redirect search without selective repair; matched evidence supports near-exact interpolation without reliable structural recovery; executable laws can compress predictions poorly; and neither law

adequacy nor complete-ranking qualification determines action validity. The oracle stop prevented a reproducible but decision-misaligned control from producing an uninterpretable participant effect. Evaluator validity is therefore part of scientific-agent validity.

7.1 A TRANSITION MAP FOR SCIENTIFIC-AGENT EVALUATION

Evaluation should therefore separate interventions from readouts, score prediction, law and action on their own units, and qualify controls on the intended endpoint. Regret, near-optimality and near-tie handling should be primary for selection, with full-rank metrics diagnostic. Evidence roles must also remain separate: the 96-query failure tests a control, the exposed 320-query pass tests construction, and the fresh failure tests generalization. Exact scheduled, eligible, completed and unstarted counts prevent these roles from being pooled into invented evidence.

7.2 LIMITATIONS AND CONCLUSION

The study concerns two models and fixed agent–tool configurations in simulation, not cross-model ranking or laboratory fidelity. C2 now has two complete 135-cell scheduled surfaces, but different failure/censoring patterns and no randomized provider assignment. A-P/B2 each contain only five worlds; DeepSeek low is not reasoning-off and its A-P block has no qualified denominator. B3 retains 13 DeepSeek schema failures, while the four-condition successor has unequal donor eligibility and substantial yoked-recipient failure. These failures remain part of the system-level estimand and preclude capability ranking or pure mediation claims. The original five-condition oracle cohort has no participant data; W2-61 has no oracle arm. Higher-fidelity artifact portability and private replication remain untested.

Scientific-agent claims should name interventions, units, failure rules, and transfer boundaries. Agency is a chain from evidence to revision, identification, compression, and decision, breakable in the agent or its evaluator.

AI USE STATEMENT

Generative AI tools assisted software implementation, experiment orchestration, data analysis, visualization code, literature discovery, and drafting and editing of the manuscript. The authors reviewed the study designs and frozen decision rules, inspected retained failures, verified reported denominators and claims against machine-readable records, and tested AI-assisted code before use. The authors take responsibility for the final content, including text, claims, code, and artifacts produced with generative-AI assistance.

REPRODUCIBILITY STATEMENT

The main paper defines the experimental units, interventions, estimands, failure handling, and stop boundaries. The appendix provides the execution and evaluation details needed to reconstruct each evidence layer. Exact action traces, evaluator-owned truth queries, replay checks, complete failure records, and machine-readable summaries will be included in the anonymous supplementary package.

ETHICS STATEMENT

The experiments use simulated chemical environments and no human participants, personal data, or live laboratory actuation. The principal risks are over-interpreting simulated results and treating an evaluator surrogate as the scientific objective; the claim boundaries and retained negative qualification results are reported to mitigate these risks.

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A APPENDIX

A.1 ENVIRONMENT SEMANTICS AND INTERVENTION CONSTRUCTION

An executable world is a stateful transition system with typed operations, measurement functions, finite resources, terminal assays, and an evaluator-only truth surface. A task publishes the operation and instrument contracts while withholding the latent state and registered relations. A scenario binds a world, public task description, resource card, and initial model. Initial-model arms alter one declared information locus without changing the transition process, measurement process, candidate actions, or evaluator truth.

The entity locus changes material identity or ontology. The parametric locus changes a quantitative response model while preserving the declared entities. The structural locus changes the functional form used to describe a mechanism. Aligned and misspecified models are matched in explicitness so that a contrast is not merely information versus no information. The opaque arm exposes the same laboratory interface but withholds the intervened model component. Exploratory, validation, prospective, and private world selections are disjoint by construction.

A.2 PERSISTENT EXECUTION AND CHECKPOINTS

Each participant cell is one persistent agent process. At every turn, the agent may call only the public laboratory tools. The host validates schemas and preconditions, applies resource accounting, commits successful transactions, and returns public observations. It does not choose or repair an experiment on the participant’s behalf. Entity, parametric, and structural campaigns permit eight, ten, and twelve complete experiments, respectively. Checkpoints occur at experiments 0/2/4/6/8, 0/2/4/7/10, and 0/3/6/9/12.

Every checkpoint records model-reliability judgements, query predictions, uncertainty, public evidence references, a typed executable law, and the intended next experiment. Checkpoint repair is part of the evaluated interface burden. All eventually yielded valid records, but rejected attempts remain in the process log and resource accounting.

A.3 EVALUATOR-OWNED PREDICTION AND LAW ASSAYS

Entity checkpoints contain four prespecified counterfactual queries; parametric and structural checkpoints contain sixteen. The evaluator executes each unique task–world query independently, never returns its truth to the participant, and shares truth across prior arms and checkpoints. The primary readout is mean normalized absolute error across registered query–metric pairs. The executed cohort contains 420/420 truth executions, 1,620 query–metric truth values, 675/675 checkpoints, 6,300 predictions, and 24,300 scored values.

Final typed laws are executed on the same registered coordinates. Schema validity, coordinate coverage, truth-normalized error, change from pre-evidence and effective-final checkpoints, and consistency with explicit final predictions are retained separately. This assay establishes executability and in-domain compression only. Reusability would require new coordinates or transfer to a context-reset process.

The matched Codex evaluator uses the identical 45-cluster, nine-task, three-arm C2 surface. DeepSeek and Codex have 121/135 and 126/135 completed cells. Overall mean prediction improvement is 0.1198/0.1329, final prediction error is 0.1685/0.1614, law MAE is 0.2371/0.1753, law compression loss is 0.0686/0.0142, and blind gain is -0.0010/-0.0001. Codex-minus-DeepSeek paired descriptive differences are +0.0131 for prediction improvement, -0.0071 for final error, -0.0648 for law MAE, and -0.0579 for compression loss. Both models fail all three registered selective-correction gates.

A.4 TYPED-LAW CAPACITY AND DISTILLATION CONTROLS

To distinguish participant compression failure from an underpowered output schema, we refitted each of the 135 complete final prediction vectors with legal identity-link laws over the registered feature coordinates and allowed basis functions. A full-basis fit used up to 64 terms per metric. A term-matched fit used exactly the participant’s submitted term budget for each metric. A leave-

one-query-out fit withheld each registered query in turn and predicted it from all remaining queries. Every fitted law was parsed and executed by the production law evaluator; independently computed design-matrix predictions had to agree with executor output within 10^{-10} .

The participant laws differed from their own final predictions by mean MAE 0.1539. Full-basis fits reproduced 135/135 final prediction states with mean MAE 4.25×10^{-13} ; term-matched fits reduced the error to 0.0114 and leave-one-query-out fits reached 0.0788. The control uses the same registered coordinate domain and therefore tests representation capacity and distillation, not global mechanistic identification or transfer.

A.5 FAILURE-AWARE INFERENCE

The unit of prospective inference is the independently selected task–world cluster, not a checkpoint, query, experiment, or replay. The three prior arms within a cluster are matched. Locus-specific selective-correction contrasts are computed first and are not pooled across different intervention semantics. Failed scientific cells remain assigned. A right-censored session carries its last valid checkpoint forward, and a missing final prediction receives zero primary improvement. Infrastructure recovery is allowed only when no scientific trajectory has persisted and only under the fixed attempt limit.

The matched-evidence structural contrast has five independent worlds. Its interval and exact one-sided sign-flip result are descriptive because the block was designed to localize a transition, not to establish a population-level arm effect. Recipe divergence is likewise a manipulation check: without repeated same-arm sessions, provider stochasticity cannot be separated from intervention sensitivity at exact-recipe resolution.

DeepSeek and GPT-5.6-sol medium used identical matched-evidence worlds, arms, evidence packets, queries, and scoring rules. Each completed 15/15 A-P and 15/15 A-S B2 formal sessions with no failed cells; canaries were excluded. A-P misspecified-minus-aligned update contrasts were 0.0309 and 0.0602, with 3/5 and 5/5 positive worlds. A-S B2 contrasts were 0.0645 and 0.0915, with 3/5 and 4/5 positive worlds; misspecified exact 1.75-law recovery was 0/5 in both models. Configurations are reported separately, and no model-superiority test is performed.

The reasoning-budget ablation kept the DeepSeek model, Codex harness, prompts, schemas, public packets, worlds, and evaluator fixed while changing only `reasoning_effort` from high to low. It is not provider-level thinking-off, which would require a different direct-controller harness. The B2 interface canary passed 3/3 and the formal block completed 15/15 sessions, 30/30 turns, 15/15 same-thread continuations, and 360/360 pre plus 360/360 post scoring terms, with no failed session or infrastructure predecessor. Mean post errors were 0.0067/0.0069/0.0069 for opaque/aligned/misspecified. The registered contrast was -0.0405, with 2/5 positive worlds, exact one-sided $p = 0.8125$, and descriptive interval [-0.1559, 0.0749]. Misspecified exact 1.75-law recovery remained 0/5; all five misspecified post errors were at most 0.02.

Provider-reported DeepSeek reasoning output was 506,637 tokens at high effort and 400,639 at low effort on the matched B2 block, a 20.9% reduction. These resource fields are descriptive within the DeepSeek provider and are not compared with GPT accounting. The separate low-effort A-P canary was platform-defective: progress events existed, but terminal cell receipts and a canary summary did not. It is retained as an unscored partial; A-P low formal execution remained 0/15 and supplies no effect estimate.

A.6 PARTICIPANT-IDENTIFIABLE LAW AND ACTION CONTROL

The independent B3 control used five frozen public structural-partition worlds, the same opaque, aligned, and misspecified prior arms, and two independent fresh GPT-5.6-sol medium sessions per arm and world. Provider-free development qualification selected eight evidence rows spanning four nominal pairs and a disjoint eight-query scoring/action roster. Public truth was executed before any participant call. All five worlds entered family, exponent, prediction, Top-1, rank, and regret denominators; action gain was scored only in the three worlds whose best candidate exceeded the visible evidence incumbent by at least 0.02. The evidence and scoring rosters, five worlds, three arms, two replicates, 0.10 exponent tolerance, and 0.02 opportunity threshold were frozen before participant execution.

A three-session canary tested only two-turn schema validity, packet identity, same-thread continuity, typed law fields, the exact scoring denominator, and selection of an unexecuted candidate. It passed 3/3 and did not enter the formal scientific denominator. The full 30-cell block then completed 30/30 with no failed cells or replacement. All sessions preserved same-thread continuity and returned 60/60 completed formal turn receipts. The canary and formal blocks together used 33 provider session attempts, 66/66 completed turn receipts, zero retries, zero tool events, and zero participant physical experiments. One formal post turn recorded a transient provider error event before returning a completed receipt; the event remains in the resource ledger and the cell remains completed.

Joint structural recovery required both the registered power family and exponent error at most 0.10. Counts for opaque, aligned, and misspecified arms were 0/10, 5/10, and 0/10. The corresponding mean post-evidence errors were 0.0367, 0.0215, and 0.0378. The aligned world-mean exponent error was lower than each comparator in all five worlds. Top-1 counts were 0/10, 2/10, and 0/10; both Top-1 choices came from one action-ineligible world. None of the 18 action-eligible cells reached gain 0.02, including neither of the two action-eligible cells with joint structural recovery. A matched DeepSeek B3 successor then started from its first cell on the same scientific surface and retained every scheduled outcome. It completed 17/30 cells; 13/30 were participant-schema failures. Failure-aware joint recovery was 0/30, Top-1 was 0/30, and mean regret was 0.9579; completed-cell post MAE was 0.0928. The corresponding Codex values were 5/30, 2/30, 0.7594, and 0.0320. Neither model achieved eligible gain at least 0.02. The matched comparison is descriptive because provider and schema failure are not randomized.

A.7 CROSS-MODEL SUCCESSOR DENOMINATORS

Historical W2-59 partials and stop boundaries remain immutable. They are not continued, pooled, or converted into successful cells. Each completion experiment instead starts at its first unit in a new root with a complete scheduled failure-aware denominator.

The Codex C2 successor scheduled all 135 task-world-prior cells and terminated with 126 completed, 3 failed, and 6 right-censored sessions, completing 1,253/1,260 participant experiments. Its provider-free evaluator completed 420/420 truth executions, scored 669/675 checkpoints and 129/135 laws, and launched 756/810 scheduled blind executions; 54 remained unstarted. The matched 135-cell DeepSeek/Codex analysis uses task-world cluster bootstrap intervals and never interprets model differences as provider effects.

The DeepSeek B3 successor similarly schedules 30 cells and retains 17 completed plus 13 participant-schema failures. Its historical canary is excluded rather than spliced into the new denominator. The Codex B3 cohort remains the independent completed 30-cell block described above.

The four-condition action successor schedules 45 strata by four conditions for each model: 180 slots per model and 360 total. DeepSeek reuses 45 immutable W2-50 autonomous donors, of which 42 are eligible; Codex creates 45 independent donors, of which 26 are eligible. No-evidence is scheduled for all strata, while yoked and learned-law recipients require the corresponding donor. Missing-donor descendants remain `not_started_due_to_missing_donor`. The primary analysis is model-specific and failure-aware; the cross-model common population contains 26 strata from 13 task-world clusters. Yoked completion is 10/42 for DeepSeek and 24/26 for Codex, so autonomous-yoked contrasts mix scientific action quality with recipient-system failure and are not pure experiment-selection effects.

A.8 COMPLETE-PLAN ACTION ASSAY

In each open-action cell, a persistent agent performs twelve autonomous experiments and commits checkpoints after 0, 3, 6, 9, and 12 experiments. The host then reveals eight unseen ActionPlans. Each plan specifies ordered operations, all submitted parameters, initial-state assumptions, intermediate measurement positions, terminal assay, and declared omitted operations. Candidate outcomes, ranks, and other-arm evidence stay hidden. The evaluator verifies identity of disclosed, truth-evaluated, and replayed plans.

The primary endpoint is normalized regret of the selected plan within its task-world candidate set. Selected rank, Top-1, pairwise ordering, and law adequacy are secondary or diagnostic. Forty-five cells were scheduled across three tasks, five worlds, and three initial-model arms. Forty-two were

eligible. The two agent-induced resource/process failures and one provider interruption are retained as failures in the scheduled denominator and are not replaced. An independent repair is reported only as sensitivity evidence because it follows a new trajectory.

The continuous law-action analysis used only those same 42 originally eligible cells; the three failures stayed in the 45-cell scheduled denominator and were not imputed. Pearson and Spearman associations were computed pooled and by task. Uncertainty resampled the 15 frozen task-world clusters, retaining all eligible arms within a sampled cluster, with 10,000 bootstrap replicates and seed 20260827. The pooled Spearman coefficients were -0.073 for law MAE versus selected rank and -0.133 for law MAE versus normalized regret, with cluster-bootstrap 95% intervals $[-0.380, 0.256]$ and $[-0.452, 0.217]$. A prespecified sensitivity table swept law-MAE thresholds 0.05, 0.075, 0.10, 0.15, 0.20, 0.25, and 0.30 without selecting a cutoff from action outcomes.

A.9 ORACLE-CONTROL QUALIFICATION

The unexecuted causal follow-up crossed three tasks, five worlds, three priors, and five conditions: no evidence, stepwise yoked evidence, autonomous exploration, learned law in a fresh context, and a provider-free oracle law. The planned denominator was 225 participant sessions, 45 autonomous donors, and 540 donor experiments. Participant execution required every fresh task-world oracle to pass candidate opportunity checks and to attain Spearman rank correlation at least 0.80 over the eight candidate plans using fit data disjoint from candidate outcomes.

The 96-query qualification stopped after eight of fifteen clusters. It completed 896/896 truth executions and exact replays, with eight candidate gates and seven oracle rank gates passing. The retained crystallization failure had $\rho = 0.738095$, Top-1 disagreement, and zero fit/candidate overlap. No operational canary or participant session began.

The 320-query construction held the fitted ExtraTrees family fixed while expanding coverage to 64 global and 256 candidate-neighborhood queries. A platform-defective partial caused by an evidence-reference schema limit was retained and not reused. After the contract repair, a new root completed 2,352/2,352 truth executions and replays across seven exposed units, passing all seven and repairing four historical failures. The first prospective world then completed 336/336 executions and replays but failed at $\rho = 0.714286$; the remaining fourteen were not started. Candidate outcomes were never read during fitting.

The gate-alignment analysis added no execution. It re-read sixteen frozen unit versions and exactly reproduced their original Spearman and Top-1 values. Its action endpoints were Top-1, normalized regret, selection within 0.01 of optimum, and near-tie-aware pair ordering. It did not alter any historical result, threshold, or stop decision.

For eight candidates without ties, $\rho = 1 - \sum_i d_i^2/84$. Thus the retained 96-query failure at $\rho = 0.738095$ has $\sum_i d_i^2 = 22$, whereas the fresh 320-query failure at $\rho = 0.714286$ has 24. The scalar measures total displacement, not whether the best action moved.

Table 2: **Complete-ranking gates and action endpoints are different estimands.**

Frozen block	n	Rank pass	Top-1	≤ 0.01 regret
Fresh 96-query	8	7	1	3
Exposed 320-query	7	7	4	6
Fresh 320-query	1	0	1	1

A.10 ADDITIONAL SCOPE BOUNDARIES

The study covers simulated chemistry and two fixed agent-tool configurations, not laboratory fidelity or universal chemical coverage. C2 and B3 have matched scheduled surfaces; A-P/B2 have complete separate denominators plus a B2-only DeepSeek-low ablation. Differential failures and non-random provider assignment prohibit model ordering. The public cohort is not private confirmation, and high-fidelity artifact portability remains untested. W2-50 lacks a same-agent no-evidence baseline; W2-61 adds four oracle-free conditions, but donor/recipient failures limit mediation claims. The original five-condition oracle cohort remains unexecuted.